

BG77xA-GL&BG95xA-GL

DFOTA Application Note

LPWA Module Series

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About the Document

Revision History

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1.0	2022-09-15	Nikola BLAZIC/ Milan DJORDJEVIC	First official release
1.1	2023-05-15	Nikola BLAZIC/ Milan DJORDJEVIC	Added the HW configuration for BG951A-GL only (Chapter 2.1).

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1 Introduction

Quectel BG77xA-GL and BG95xA-GL modules support DFOTA (Delta Firmware Upgrade Over-The-Air) feature, which allows users to update firmware wirelessly.

A delta firmware package, which contains only the differences between the current and the target update version, is needed before updating firmware in DFOTA. Therefore, DFOTA is time-saving and can reduce unnecessary data transmission.

This document mainly explains how to update the firmware of BG77xA-GL and BG95xA-GL modules via DFOTA, which can be triggered either by **AT+QFOTADL** or the LwM2M server.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Model
BG77xA-GL	BG770A-GL
	BG773A-GL
BG95xA-GL	BG950A-GL
	BG951A-GL
	BG953A-GL
	BG955A-GL

1.2. Special Mark

Table 2: Special Mark

Mark	Definition
*	Unless otherwise specified, an asterisk (*) after a function, feature, interface, pin name, command, or argument indicates that the function, feature, interface, pin, AT command, or argument is under development and currently not supported; and the asterisk (*) after a model indicates that the sample of the model is currently unavailable.

2 Trigger DFOTA by AT Commands

2.1. Prerequisite

The following HW configuration needs to be applied for BG951A-GL module only:

- Drive GNSS_BOOT low or keep it floating.
- Drive GNSS_EN low or keep it floating.
- Perform module reset.

2.2. Firmware Update Procedure

The following chart illustrates the DFOTA procedure triggered by the AT command, provided that the delta firmware package is stored on an HTTP(S)/CoAP(S) server.

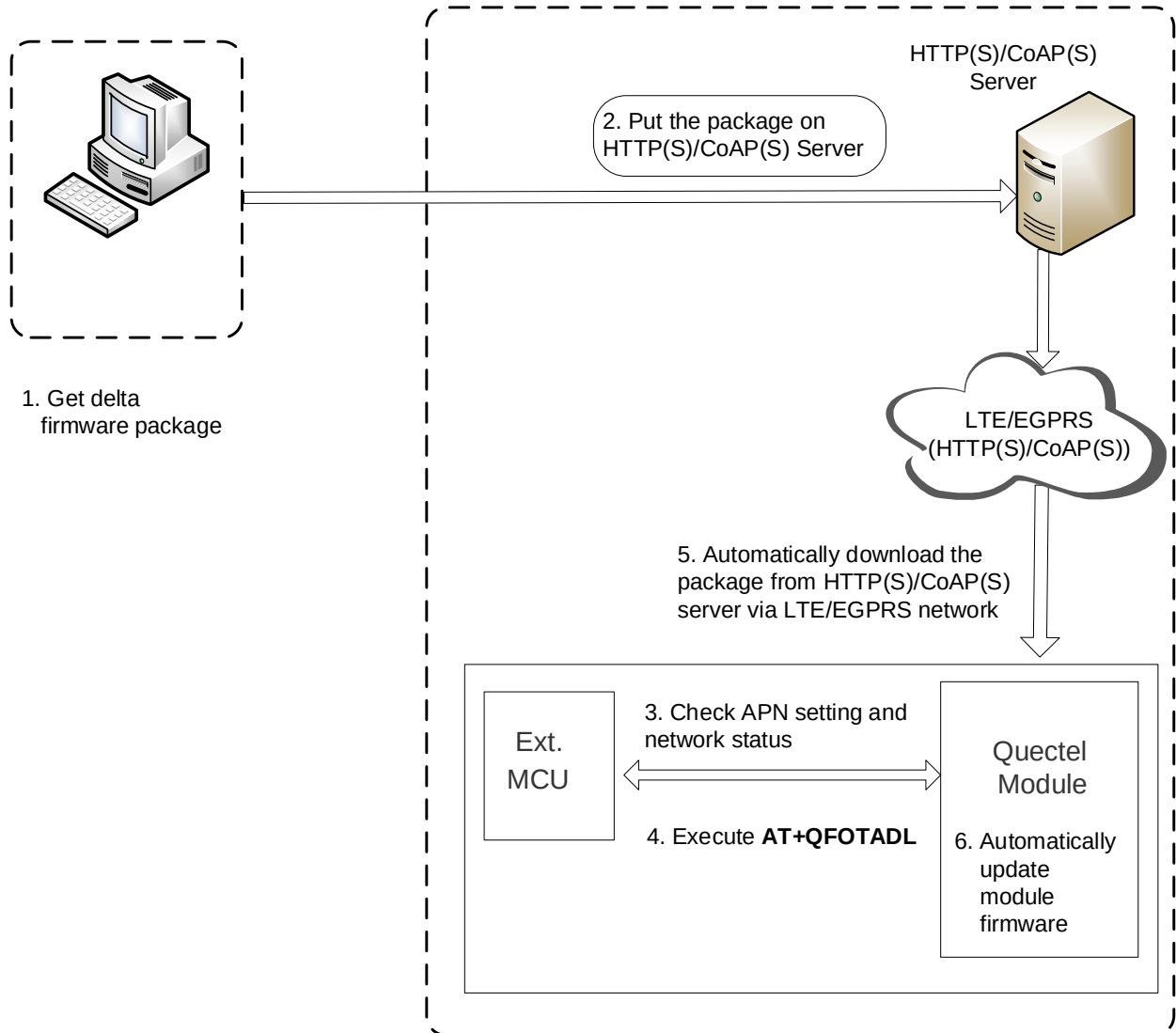


Figure 1: DFOTA Procedure (Triggered by AT Command)

As shown in the above figure, the following steps need to be performed to update the firmware when the delta firmware package is stored on the HTTP(S)/CoAP(S) server:

- Step 1:** Get a delta firmware package from Quectel.
- Step 2:** Put the delta firmware package on the HTTP(S)/CoAP(S) server.
- Step 3:** Check the APN setting and network status.
- Step 4:** Execute **AT+QFOTADL**, and the delta firmware package will be downloaded automatically to the module from the HTTP(S)/CoAP(S) server via LTE/EGPRS network (**Step "5"** in the figure).

above). Then, the firmware of the module will be updated automatically (**Step "6"** in the figure above).

2.2.1. Get Delta Firmware Package

Before updating, check the current firmware version with **AT+GMR** and select the target firmware version, and then send the two firmware versions to Quectel or module supplier to get a delta firmware package.

2.2.2. Put Delta Firmware Package on HTTP(S)/CoAP(S) Server

Step 1: Set up the HTTP(S)/CoAP(S) server before using the DFOTA function (Quectel does not provide such servers).

Step 2: Put the delta firmware package on the server and record its storage path.

Step 3: After executing the corresponding AT command, the module downloads the delta firmware package using the URL path of the file uploaded on the server.

2.2.3. Check APN Setting and Network Status

Step 1: Check APN setting: check if the APN is set properly after putting the delta firmware package on the HTTP(S)/CoAP(S) server. For instance, the APN for a Telstra (U)SIM card should be "Telstra.internet". The relevant AT commands are listed below:

- **AT+QCFGEXT="fota_apn"**: Query DFOTA APN.
- **AT+QCFGEXT="fota_apn",0,"Telstra.internet"**: Set DFOTA APN.

Step 2: Check network status: after the APN setting is confirmed, make sure that the data network is registered before firmware updating. The relevant AT commands are listed below:

- **AT+CSQ**: Query signal quality
- **AT+CEREG?**: Query network registration status (Optional: **AT+CGREG?** for modules using 2G network)
- **AT+COPS?**: Query the registered operator

For more information about the above commands, see **document [1]**.

2.2.4. Execute AT Command to Update Firmware

After checking the APN setting and network status, execute **AT+QFOTADL** to download the delta firmware package to the module from the HTTP(S)/CoAP(S) server via LTE/EGPRS network. Then, the firmware of the module will be updated. For an example, see **Chapter 2.4**.

2.3. DFOTA-Related AT Commands

2.3.1. AT Command Introduction

2.3.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.3.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 3: Type of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

2.3.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2.3.3. Description of AT Command Mode and Data Mode

The COM port of the modules has two working modes: AT command mode and data mode. In AT command mode, the data inputted via the COM port are treated as AT commands, while in data mode, they are treated as data.

● Exit Data Mode

Inputting **+++** or pulling the MAIN_DTR pin up can make the COM port exit data mode. To prevent **+++** from being misinterpreted as data:

- 1) Do not input any character for at least 1 s before and after inputting **+++**.
- 2) Input **+++** within 1 s, and wait until **OK** is returned. When **OK** is returned, the COM port exits data mode and enters AT command mode.

If you are exiting data mode by pulling MAIN_DTR up, make sure to set **AT&D1** first.

● Enter Data Mode

After **AT+QFOTADL="COM:",<file_size>,<timeout>** is executed, the COM port enters data mode. If you input **+++** or pull the MAIN_DTR pin up to make the port exit data mode, the execution of this command will be interrupted before a response is returned. In such a case, the COM port cannot re-enter data mode if you execute **ATO**.

NOTE

See *document [1]* for more information on **AT&D** and **ATO**.

2.3.4. AT+QFOTADL Update Firmware via DOFTA

AT+QFOTADL Update Firmware via DOFTA

Test Command	Response
AT+QFOTADL=?	OK

NOTE

Before downloading, via HTTP(S), CoAP(S) or COM port, the UFS partition will be erased.

2.3.4.1. AT+QFOTADL=<HTTP_URL> Update Firmware When Delta Firmware Package is Stored on HTTP(S) Server

This command initiates automatic module firmware updating via DFOTA. Once **AT+QFOTADL** is executed, the delta firmware package is automatically downloaded to the module. After downloading the delta firmware package, the module automatically updates the firmware. The module reboots if the firmware has been updated successfully, otherwise, it returns an error message and exits DFOTA.

If the delta firmware package is stored on the HTTP(S) server, **AT+QFOTADL=<HTTP_URL>** is executed to start automatic firmware updating via DFOTA. The module downloads the delta firmware package from the HTTP(S) server over the air and updates the firmware automatically.

AT+QFOTADL=<HTTP_URL> Update Firmware When Delta Firmware Package is Stored on HTTP(S) Server

Write Command	Response
AT+QFOTADL=<HTTP_URL>	OK
	+QIND: "FOTA","HTTPSTART"
	+QIND: "FOTA","DOWNLOADING",<percent>
	+QIND: "FOTA","DOWNLOADING",<percent>
	...
	+QIND: "FOTA","HTTPEnd",<HTTP_err>
	+QIND: "FOTA","RESETTING"
	+QIND: "FOTA","START"
	+QIND: "FOTA","UPDATING",<percent>
	+QIND: "FOTA","UPDATING",<percent>
	...
	+QIND: "FOTA","END",<update_err>
	If there is any error:
	+CME ERROR: <err>

Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<HTTP_URL>	String type. Maximum length: 700 bytes. It should start with ""HTTP://" or "HTTPS://". For example: "HTTP://<HTTP_server_URL>:<HTTP_port>/<HTTP_file_path>"
<HTTP_server_URL>	String type. IP address or domain name of the HTTP(S) server.
<HTTP_port>	Integer type. HTTP(S) server port number. Default value: 80. Range: 1-65535.
<HTTP_file_path>	String type. File path on HTTP(S) server.
<HTTP_err>	Integer type. HTTP(S) result code. 0 Downloaded successfully. Any other value Error code. See Chapter 4 for more information.
<percent>	Integer type. Download progress is expressed as a percentage.
<update_err>	Integer type. Update result code. 0 Updated successfully. Any other value Error code. See Chapter 4 for more information.
<err>	Integer type. Error code. See Chapter 4 for more information.

NOTE

1. In the DFOTA mode, there will be no response on AT port until updating is finished. The updating might last for a few minutes.
2. After a smooth downloading process, the module is reset to enter the DFOTA mode. Once the updating is finished, the module restarts into the normal mode.
3. If the module is powered down during the updating, the module automatically enters the DFOTA mode and continues the interrupted firmware updating the next time it is powered up.
4. 1) If SSL certificates are needed for downloading the firmware package from an HTTPS server, they should be loaded with **AT+QHTTPCFG="sslctxid",<sslctxID>**. For more information, see **document [2]**.
2) Certificate uploading method: **AT+QFUPL** can be used to upload the certificates. For more information about the command, see **document [3]**.

2.3.4.2. AT+QFOTADL=<CoAP_URL> Update Firmware When Delta Firmware Package is Stored on CoAP(S) Server

If the delta firmware package is stored on a CoAP(S) server, execute this command to enable automatic firmware updating via DFOTA. The module downloads the delta firmware package from the CoAP(S) server wirelessly and updates the firmware automatically.

AT+QFOTADL=<CoAP_URL> Update Firmware When Delta Firmware Package is Stored on CoAP(S) Server

Write Command AT+QFOTADL=<CoAP_URL>	Response OK +QIND: "FOTA","COAPSTART" +QIND: "FOTA","COAPEND",<CoAP_err> +QIND: "FOTA","RESETTING" +QIND: "FOTA","START" +QIND: "FOTA","UPDATING",<percent> +QIND: "FOTA","UPDATING",<percent> ... +QIND: "FOTA","END",<update_err> If there is any error: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<CoAP_URL>	String type. Maximum length: 700 bytes. It should start with " CoAP://" or "CoAPS://". For example: "CoAP://<CoAP_server_URL>:<CoAP_port>/<CoAP_file_path>"
<CoAP_server_URL>	String type. IP address or domain name of the CoAP(S) server.
<CoAP_port>	Integer type. CoAP(S) server port number. Default value: 5683. Range: 1-65535.
<CoAP_file_path>	String type. File path on CoAP(S) server.
<CoAP_err>	Integer type. CoAP(S) result code. 0 Downloaded successfully. Any other value Error code. See Chapter 4 for more information.
<update_err>	Integer type. Update result code. 0 Updated successfully. Any other value Error code. See Chapter 4 for more information.

<err> Integer type. Error code. See **Chapter 4** for more information.

NOTE

If SSL certificates are needed for downloading the delta firmware package from a CoAP server, they should be loaded with **AT+QCOAPCFG="dtls",<clientID>[<DTLS_enable>,<DTLS_contextID>]]**. For more information, see **document [4]**.

2.3.4.3. AT+QFOTADL="COM:",<file_size> Upload a Delta Firmware Package to Flash

This command uploads a delta firmware package to the NAND flash. If a delta firmware package already exists in the NAND flash, it will be overwritten when a new package is uploaded.

After this command is executed and **CONNECT** is returned, the module switches to data mode. When the uploaded data reach <file_size>, or if no data are inputted when <timeout> is reached, the module exits data mode automatically. During data transmission, inputting +++ or pulling up MAIN_DTR can make the module exit data mode. More information about data mode is provided in **Chapter 2.3.3**.

AT+QFOTADL="COM:",<file_size> Upload a Delta Firmware Package to Flash

Write Command AT+QFOTADL="COM:",<file_size>[,<timeout>]	Response CONNECT TA switches to data mode (transparent transmission mode), and the binary delta firmware package can be inputted. When the total size of the inputted data reaches <file_size>, TA returns to command mode and sends the following response: +QFOTADL: <upload_size>,<check_sum> OK If there is any error: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<file_size>	Integer type. File size expected to be uploaded. Unit: byte.
<timeout>	Integer type. Waiting time for data to be inputted to UART. Default value: 5. Range: 1–65535. Unit: s.

<upload_size>	Integer type. Uploaded file size. Unit: byte.
<check_sum>	Checksum of uploaded data.
<err>	Integer type. Error code. See Chapter 4 for more information.

NOTE

1. <file_size> is limited by the flash space for the delta firmware package. The maximum flash space of the modules is 3788 KB.
2. <check_sum> is a 16-bit checksum based on bitwise Exclusive-OR (XOR). It is calculated by XOR-ing all even bytes (stored as upper 8 bits), and likewise XOR-ing all odd bytes (stored as lower 8 bits) of this checksum. This checksum can be used to verify if the data have been uploaded correctly.
3. +++ sequence causes TA to end command execution and switch to command mode. However, the previously uploaded data are preserved into the file.
4. When executing the command, the data must be entered after **CONNECT** is returned.
5. Due to the physical limitations of the platform, i.e., write speed to the NOR flash memory, the upload feature has 2 operating modes to guarantee that the entire file will be written without data loss. The two modes are: 1) uploading without activated flow control; 2) uploading with activated flow control. If the flow control is not activated, the maximum file size is limited to 200,000 bytes. Attempting to send larger files will result in an error. When HW flow is activated by executing **AT+IFC=2,2**, there is no file size limitation other than available disk space. In this mode, on the client-side, the RTS and CTS must be switched on before sending the file to ensure safe file transfer. If RTS and CTS pins are not physically connected, then the software flow control is available as an alternative. The software flow control can be enabled using **AT+QCFGEXT="xonoff",1**. This implementation of SW flow control is only implemented for data. This means that the user must listen for the Xon (0x11) and Xoff (0x13) and interpret them as the flow control characters. In this case, when sending data there are no special characters.

2.3.4.4. AT+QFOTADL=<para1>[,<para2>] Extended DFOTA Functions

This command supports several extended functions with different parameters, such as: triggering a DFOTA update with a local DFOTA firmware package, getting DFOTA firmware package information, and deleting a DFOTA firmware package.

AT+QFOTADL=<para1>[,<para2>] Extended DFOTA Functions

Write Command	Response
AT+QFOTADL=<para1>[,<para2>]	If <para2> is omitted and <para1>=1, start DFOTA updating. OK
	If <para2> is omitted and <para1>=3, return the current DFOTA update trigger mode. +QFOTADL: <para1>,<para2>

	OK
	If <para2> is omitted and <para1>=4 , return the size of the local DFOTA package. +QFOTADL: <para1>,<para2>
	OK
	If <para2> is omitted and <para1>=5 , delete the DFOTA package. OK
	If <para2> is present and <para1>=3 , configure the DFOTA update trigger mode. OK
	If there is any error related to ME functionality: +CME ERROR: <err>
	If there is any other error: ERROR
Maximum Response Time	10 s
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<para1>	Integer type. 1 Start DFOTA updating 2 Reserved 3 Configure whether to trigger DFOTA updating automatically after downloading the delta firmware package (takes effect on HTTP(S) and CoAP(S)) 4 Get the size of local delta firmware package 5 Delete the local delta firmware package
<para2>	If <para1>=3 , 0 Trigger DFOTA updating with AT+QFOTADL=1 . 1 Trigger DFOTA updating automatically once the delta firmware package is downloaded. If <para1>=4 , <para2> represents the size of the local delta firmware package. Unit: Byte.
<err>	Integer type. Error code. See Chapter 4 for more information.

2.3.5. AT+QCFGEXT="fota_apn" Configure IP Family and APN for DFOTA*

This command configures the IP family and APN for DFOTA.

AT+QCFGEXT="fota_apn" Configure IP Family and APN for DFOTA

Write Command AT+QCFGEXT="fota_apn",<iptype>,<apn>[,<username>,<password>]	Response If the optional parameters are omitted, query the current setting: +QCFGEXT: "fota_apn",<iptype>,<apn>[,<username>,<password>] OK If any of the optional parameters is specified, set the IP family and APN for DFOTA: OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<iptype>	Integer type. IP family. 0 IPv4 address family 1 IPv6 address family 2 IPv4 and IPv6 address family
<apn>	String type. Access point name.
<username>	String type. Username of selected APN.
<password>	String type. Password of selected APN.

2.4. Example

Update the firmware when the delta firmware package is stored on an HTTPS server.

```
//The delta firmware package has already been stored on an HTTPS server.
//The HTTPS server address is "https://www.quectel.com:100/update.ua".
AT+CSQ;+CEREG?;+COPS? //Query network status.
+CSQ: 26,99
```

+CEREG: 0,1

+COPS: 0,0,"Telstra Mobile Telstra",7

OK

//Execute **AT+QFOTADL** to enable automatic firmware updating via DFOTA, and then the module starts to download the delta firmware package and update firmware automatically.

AT+QFOTADL="https://www.quectel.com:100/update.ua"

OK

+QIND: "FOTA","HTTPSTART"

+QIND: "FOTA","DOWNLOADING",5%

...

+QIND: "FOTA","DOWNLOADING",100%

+QIND: "FOTA","HTTPEND",0

//Finish downloading the delta firmware package from the HTTPS server.

+QIND: "FOTA","RESETTING"

//The module is reset and then enters DFOTA mode.

+QIND: "FOTA","START"

+QIND: "FOTA","UPDATING",1%

+QIND: "FOTA","UPDATING",8%

...

+QIND: "FOTA","UPDATING",100%

+QIND: "FOTA","END",0

//Finish updating the firmware.

3 Trigger DFOTA with LwM2M*

3.1. Firmware Update Procedure

OMA defines the application layer communication protocol between an LwM2M server and an LwM2M client, which is located in an LwM2M device. For the modules, the LwM2M protocol is used to manage device provisioning and triggers a DFOTA procedure. For the DFOTA update on the module, a Firmware Update Object is provided by the LwM2M specification.

Routine DFOTA updating with LwM2M steps:

- Step 1:** Write "Package"/"Package URI" to trigger the downloading of a firmware image, so that the DFOTA application automatically downloads the firmware image from an LwM2M server.
- Step 2:** Post "Update" to trigger firmware updating on the module, so that the DFOTA application automatically finishes firmware updating with the candidate firmware image.

The LwM2M client is notified about the new firmware by the LwM2M server using CoAP(S). The client sends the notification to a registered DFOTA application. The DFOTA application downloads the delta firmware package from the server and stores it in the file system of the modules.

Key components of the LwM2M DFOTA process on the module are presented in the following figure.

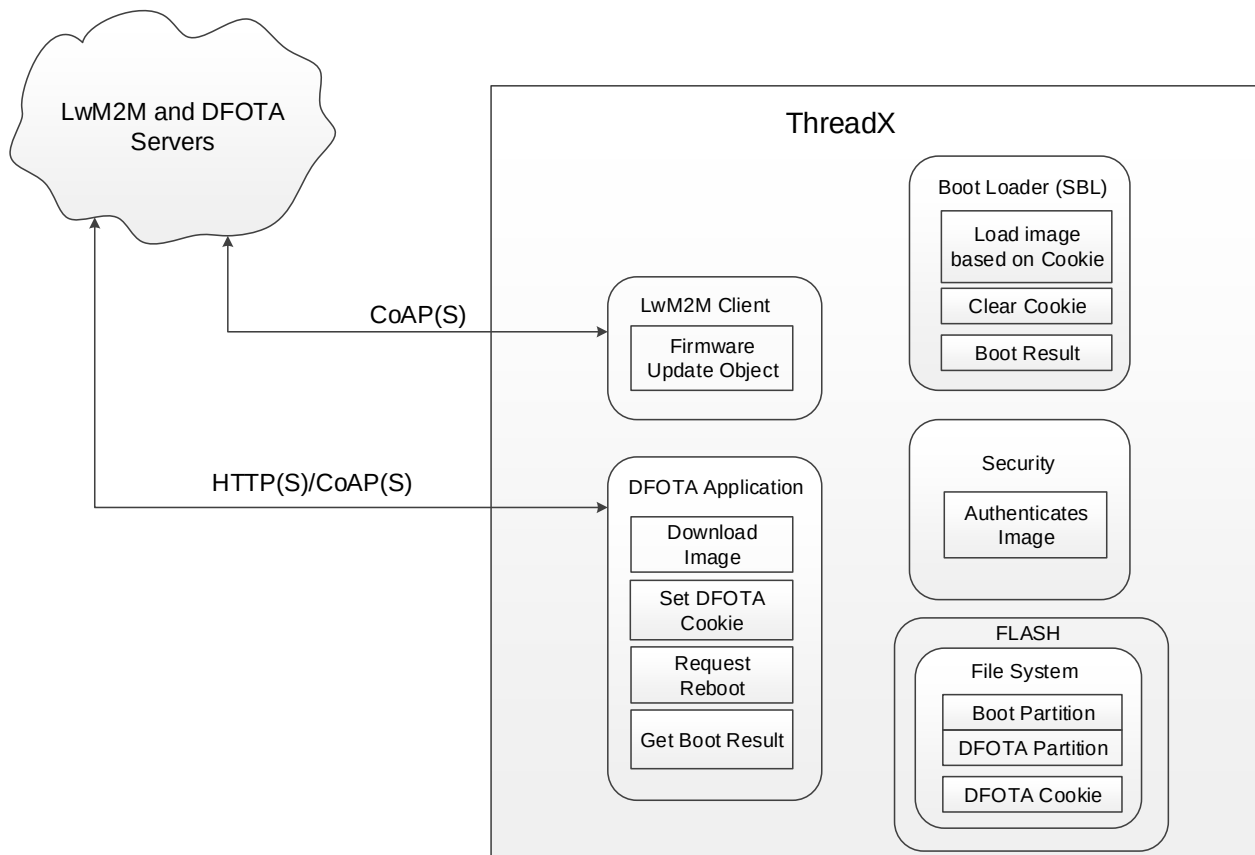


Figure 2: LwM2M Firmware Image Downloading

3.1.1. Download Firmware Image

About "Firmware Update Object" in LwM2M, the detailed interaction between an LwM2M client and an LwM2M server is described below.

The LwM2M client should register and observe the "Package URI" resource in the Firmware Update Object, which is shown on the server when the client is registered successfully.

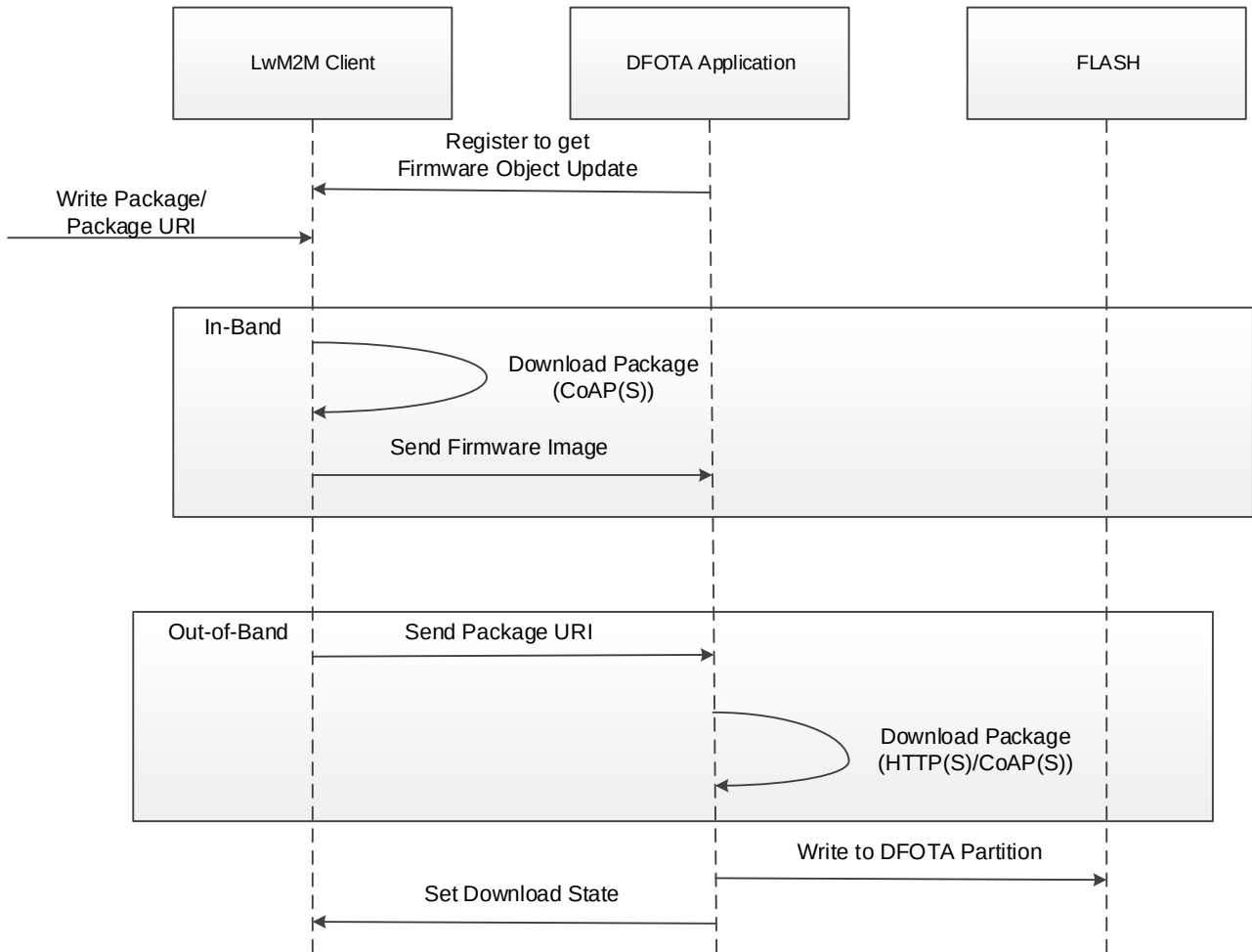


Figure 3: Call Flow of Package URI Observing and Firmware Downloading

NOTE

In the In-Band mode, the module downloads a firmware package from the LwM2M server directly via the CoAP(S) protocol and sends the downloaded package to the DFOTA application. All "firmware update" actions are done by the application. However, in the Out-of-Band mode, the module downloads a firmware package from the HTTP(S)/CoAP(S) server to the module.

3.1.2. Update Firmware

The LwM2M server should observe the "State" resource in the Firmware Update Object. If the "State" has changed to "Downloaded", "Update" resource is executed to trigger the firmware updating procedure.

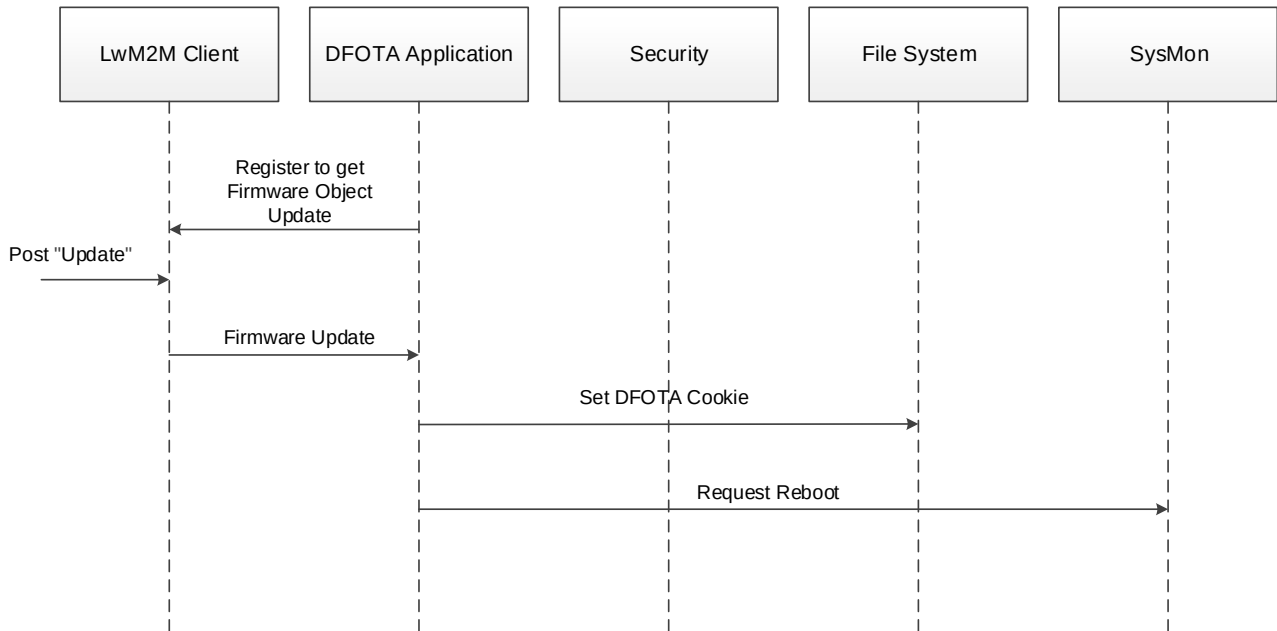


Figure 4: Call Flow of "Update" Resource Observing and Update Event Processing

3.2. URC of Firmware Updating with Lwm2M

3.2.1. Download Progress URCs

3.2.1.1. In-Band Mode

When the DFOTA firmware updating is triggered in the Lwm2M in-band mode, the following URCs are outputted to notify the user of downloading progress.

+LWM2M: "FOTA","DOWNLOADING" Notify Downloading Progress

URC Format:	Downloading procedure started.
+LWM2M: "FOTA","DOWNLOADING","START"	
URC Format:	Downloading procedure completed.
+LWM2M: "FOTA","DOWNLOADING","COMPLETED"	

3.2.1.2. Out-of-Band HTTP(S) Mode

When the DFOTA firmware updating is triggered in the Lwm2M out-of-band HTTP(S) mode, the following URCs are outputted to notify the user of the downloading progress.

+QIND: "FOTA","HTTP" Notify Downloading Progress

URC Format: +QIND: "FOTA","HTTPSTART"	Downloading procedure started.
URC Format: +QIND: "FOTA","DOWNLOADING",<percent> ...	Downloading progress in percentage.
URC Format: +LWM2M: "FOTA","DOWNLOADED",<HTTP_err>	Downloading operation result.

Parameter

<percent>	Integer type. Downloading progress in percentage.
<HTTP_err>	Integer type. HTTP(S) result code.
0	Downloaded successfully.
Any other value	Error code. See Chapter 4 for more information.

3.2.1.3. Out-of-Band CoAP(S) Mode

When the DFOTA firmware update is triggered in the LwM2M out-of-band CoAP(S) mode, the following URCs are outputted to notify the user of downloading progress.

+QIND: "FOTA","COAP" Notify Downloading Progress

URC Format: +QIND: "FOTA","COAPSTART"	Downloading procedure started.
URC Format: +LWM2M: "FOTA","DOWNLOADED",<CoAP_err>	Downloading operation result.

Parameter

<CoAP_err>	Integer type. CoAP(S) result code.
0	Downloaded successfully.
Any other value	Error code. See Chapter 4 for more information.

3.2.2. Update Progress URCs

3.2.2.1. +QIND: "FOTA","START" Updating Started

+QIND: "FOTA","START" Updating Started

URC Format:

+QIND: "FOTA","START"

Notify the user of update start.

3.2.2.2. +QIND: "FOTA","END",<err> Updating Finished

+QIND: "FOTA","END",<err> Updating Finished

URC Format:

+QIND: "FOTA","END",<err>

Notify the user of update completion.

Parameter

<err> Error code. See **Chapter 4** for more information.

4 Summary of Error Codes

These error codes indicate errors related to mobile equipment or network. The detailed information about `<HTTP_err>`, `<CoAP_err>`, `<update_err>` and `<err>` is provided in the following tables.

Table 4: Summary of `<HTTP_err>/<CoAP_err>` Codes

<code><HTTP_err>/<CoAP_err></code>	Meaning
0	Download successful
701	Unknown error
702	Server connection failed
703	Request failed
704	Download timeout
705	URL error
706	File does not exist
707	Writing data to file failed
708	Downloaded file is too large
710	Downloaded package check failed

Table 5: Summary of `<update_err>` Codes

<code><update_err></code>	Meaning
0	Updated successfully
504	Firmware update failed
505	Update package does not exist

506	Package data integrity check failed
511	Package is mismatched with the current firmware
512	DFOTA unknown error

Table 6: Summary of <err> Codes

<err>	Meaning
421	Time-out
427	Failed to allocate memory. If file size is > 200 KB then the flow control must be used
590	Input parameter invalid
591	DFOTA is ongoing

5 Appendix References

Table 7: Related Documents

Document Name
[1] Quectel_BG77xA-GL&BG95xA-GL_AT_Commands_Manual
[2] Quectel_BG770A-GL&BG95xA-GL_HTTP(S)_Application_Note
[3] Quectel_BG770A-GL&BG95xA-GL_FILE_Application_Note
[4] Quectel_BG77xA-GL&BG95xA-GL_CoAP_Application_Note

Table 8: Terms and Abbreviations

Abbreviation	Description
AP	Application Processor
APN	Access Point Name
APP	Application
CoAP(S)	Constrained Application Protocol (Secure)
CTS	Clear to Send
DFOTA	Delta Firmware Upgrade Over-the-Air
DTLS	Datagram Transport Layer Security
EFS	Embedded File System
EGPRS	Enhanced General Packet Radio Service
HTTP(S)	Hyper Text Transport Protocol (Secure)
HW	Hardware

IP	Internet Protocol
IPv4	Internet Protocol Version 4
IPv6	Internet Protocol Version 6
KB	Kilobytes
LTE	Long-Term Evolution
LPWA	Low-Power Wide-Area
LwM2M	Lightweight M2M
MCU	Microcontroller Unit/Microprogrammed Control Unit
NAND	NOT-AND
NOR	NOT-OR
OMA	Open Mobile Alliance
PSK	Pre-Shared Key/Phase Shift Keying
RTS	Ready to Send
SBL	Serial Bootloader
SSL	Secure Sockets Layer
TA	Terminal Adapter
UART	Universal Asynchronous Receiver/Transmitter
URC	Unsolicited Result Code
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
USB	Universal Serial Bus
(U)SIM	Universal Subscriber Identity Module
XOR	Exclusive-OR